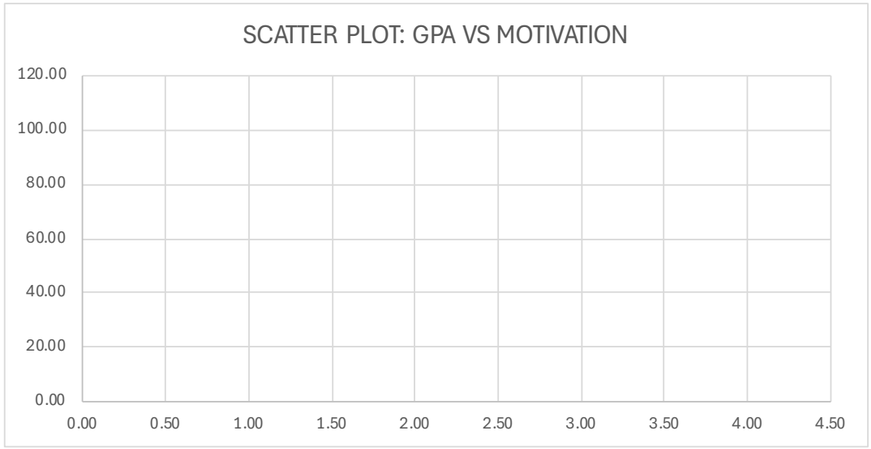
PEARSON CORRELATION COEFFICIENT — GPA & MOTIVATION

Manual calculation using summary statistics

1. DATA AND CALCULATIONS

GPA (X)	MOTIVATION (Y)	X²	Y²	XY
2.00	50.00	4.00	2500.00	100.00
2.00	48.00	4.00	2304.00	96.00
2.00	100.00	4.00	10000.00	200.00
2.00	12.00	4.00	144.00	24.00
2.30	34.00	5.29	1156.00	78.20
2.60	30.00	6.76	900.00	78.00
2.60	78.00	6.76	6084.00	202.80
3.00	87.00	9.00	7569.00	261.00
3.10	84.00	9.61	7056.00	260.40
3.20	75.00	10.24	5625.00	240.00
3.60	83.00	12.96	6889.00	298.80
3.80	90.00	14.44	8100.00	342.00
3.80	90.00	14.44	8100.00	342.00
4.00	98.00	16.00	9604.00	392.00
40.00	959.00	121.50	76031.00	2915.20

2. SCATTER PLOT



3. SUMMARY STATISTICS

SUM (X)	40.00
SUM (Y)	959.00
SUM (X²)	121.50
SUM (Y²)	76031.00
SUM (XY)	2915.20
n	14.00

4. PEARSON CORRELATION CALCULATION

SSxx	7.214285714	SSxx = Σx² − (Σx)²/n
SSyy	10339.500000000	SSyy = Σy² − (Σy)²/n
SSxy	175.200000000	SSxy = Σxy − (Σx × Σy)/n
Pearson's r — formula	0.641486711	r = SSxy / √(SSxx × SSyy)
Pearson's r — Excel function	0.641486711	CORREL(X range, Y range)

5. INTERPRETATION

Pearson correlation coefficient, r = 0.6415. This indicates a moderately strong positive linear relationship between GPA and MOTIVATION: students with higher GPA values tend to have higher motivation scores in this dataset. Correlation alone does not establish causation.

6. R CODE

```
GPA <- c(2, 2, 2, 2, 2.3, 2.6, 2.6, 3, 3.1, 3.2, 3.6, 3.8, 3.8, 4)

MOTIVATION <- c(50, 48, 100, 12, 34, 30, 78, 87, 84, 75, 83, 90, 90, 98)

cor(GPA, MOTIVATION, method = "pearson")

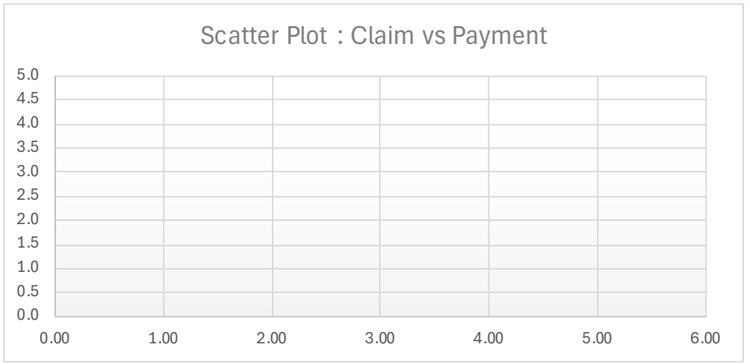
cor.test(GPA, MOTIVATION, method = "pearson")
```

PEARSON CORRELATION — CLAIM & PAYMENT

1. DATA AND CALCULATIONS

CLAIM (X)	PAYMENT (Y)	X^2	Y^2	XY
2.10	2.1800	4.41	4.7524	4.5780
2.40	2.0600	5.76	4.2436	4.9440
2.50	2.5400	6.25	6.4516	6.3500
3.20	2.6100	10.24	6.8121	8.3520
3.60	3.6700	12.96	13.4689	13.2120
3.80	3.2500	14.44	10.5625	12.3500
4.10	4.0200	16.81	16.1604	16.4820
4.20	3.7100	17.64	13.7641	15.5820
4.50	4.3800	20.25	19.1844	19.7100
5.00	4.4500	25.00	19.8025	22.2500
35.40	**32.8700**	**133.76**	**115.2025**	**123.8100**

2. SCATTER PLOT



3. SUMMARY STATISTICS

SUM (X)	35.40
SUM (Y)	32.8700
SUM (X^2)	133.76
SUM (Y^2)	115.2025
SUM (XY)	123.8100
n	10

4. PEARSON CORRELATION CALCULATION

SSxx	8.444000000	$SS_{xx} = \Sigma x^2 - (\Sigma x)^2 / n$
SSyy	7.158810000	$SS_{yy} = \Sigma y^2 - (\Sigma y)^2 / n$
SSxy	7.450200000	$SS_{xy} = \Sigma xy - (\Sigma x \times \Sigma y) / n$
Pearson's r — formula	0.958238026	$SSxy / \sqrt{(SSxx \times SSyy)}$
Pearson's r — Excel function	0.958238026	*CORREL(X range, Y range)*

5. INTERPRETATION

Pearson correlation coefficient, r = 0.9582. This indicates a very strong positive linear relationship between CLAIM and PAYMENT: higher CLAIM values tend to be associated with higher PAYMENT values. Correlation alone does not establish causation.

6. R CODE

```
claim <- c(2.1, 2.4, 2.5, 3.2, 3.6, 3.8, 4.1, 4.2, 4.5, 5.0)
payment <- c(2.18, 2.06, 2.54, 2.61, 3.67, 3.25, 4.02, 3.71, 4.38, 4.45)

cor(claim, payment, method = "pearson")
cor.test(claim, payment, method = "pearson")
```

CORRELATION ANALYSIS — LEVERAGE RATIO & INTEREST RATE

Comparison before and after log transformation of the response variable

1. ORIGINAL DATA (BEFORE TRANSFORMATION)

Leverage Ratio (X)	Interest Rate (Y)	X^2	Y^2	XY
0.10	2.80	0.01	7.84	0.28
0.40	3.40	0.16	11.56	1.36
0.50	3.50	0.25	12.25	1.75
0.80	3.60	0.64	12.96	2.88
1.00	4.60	1.00	21.16	4.60
1.80	6.30	3.24	39.69	11.34
2.00	10.20	4.00	104.04	20.40
2.50	19.70	6.25	388.09	49.25
2.80	31.30	7.84	979.69	87.64
3.00	42.90	9.00	1840.41	128.70
14.90	**128.30**	**32.39**	**3417.69**	**308.20**

2. TRANSFORMED DATA (LN INTEREST RATE)

Leverage Ratio (X)	LN Interest Rate (Y*)	X^2	$(Y^*)^2$	X × Y*
0.10	1.03	0.01	1.06	0.1030
0.40	1.22	0.16	1.50	0.4895
0.50	1.25	0.25	1.57	0.6264
0.80	1.28	0.64	1.64	1.0247
1.00	1.53	1.00	2.33	1.5261
1.80	1.84	3.24	3.39	3.3130
2.00	2.32	4.00	5.39	4.6448
2.50	2.98	6.25	8.88	7.4515
2.80	3.44	7.84	11.86	9.6421
3.00	3.76	9.00	14.13	11.2766
14.90	**20.66**	**32.39**	**51.75**	**40.10**

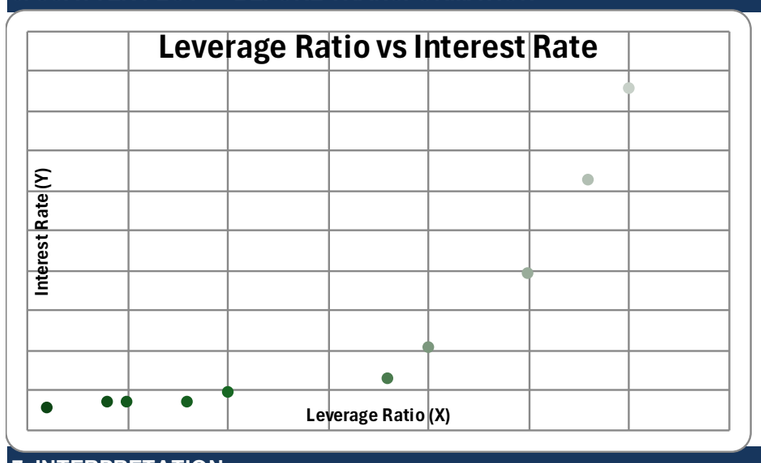
3. SUMMARY STATISTICS — ORIGINAL

SUM X	14.90
SUM Y	128.30
SUM X^2	32.39
SUM Y^2	3417.69
SUM XY	308.20
n	10
SSxx	10.19
SSyy	1771.60
SSxy	117.03
Pearson r	0.8711

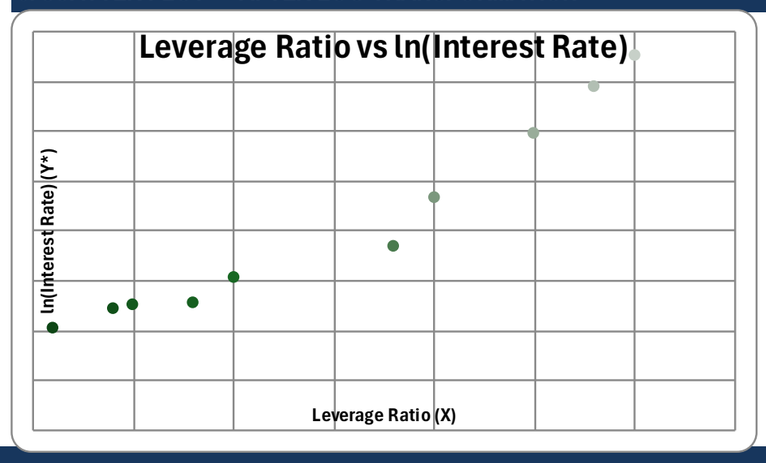
4. SUMMARY STATISTICS — TRANSFORMED

SUM X	14.90
SUM Y*	20.66
SUM X^2	32.39
SUM $(Y^*)^2$	51.75
SUM X·Y*	40.10
n	10
SSxx	10.19
SSyy*	9.07
SSxy*	9.32
Pearson r	0.9691

5. SCATTER PLOT — BEFORE TRANSFORMATION



6. SCATTER PLOT — AFTER LOG TRANSFORMATION



7. INTERPRETATION

Before transformation: Pearson r = 0.8711, indicating a very strong positive linear association between leverage ratio and interest rate. The original scatter plot may show curvature because interest rate rises more rapidly at larger leverage ratios.

After transformation: Pearson r = 0.9691, indicating a very strong positive linear association between leverage ratio and ln(interest rate). Compare the two scatter plots to assess whether the log transformation makes the relationship more nearly linear. A

8. R CODE

CORRELATION ANALYSIS — LEVERAGE RATIO & INTEREST RATE

Comparison before and after log transformation of the response variable

1. ORIGINAL DATA (BEFORE TRANSFORMATION)

Leverage Ratio (X)	Interest Rate (Y)	X²	Y²	XY

2. TRANSFORMED DATA (LN INTEREST RATE)

Leverage Ratio (X)	LN Interest Rate (Y*)	X²	(Y*)²	X × Y*

```
leverage <- c(0.1, 0.4, 0.5, 0.8, 1.0, 1.8, 2.0, 2.5, 2.8, 3.0)
interest_rate <- c(2.8, 3.4, 3.5, 3.6, 4.6, 6.3, 10.2, 19.7, 31.3, 42.9)

# Before transformation
plot(leverage, interest_rate, main = "Before Transformation",
     xlab = "Leverage Ratio", ylab = "Interest Rate", pch = 19)
abline(lm(interest_rate ~ leverage), col = "blue", lwd = 2)
cor(leverage, interest_rate, method = "pearson")
cor.test(leverage, interest_rate, method = "pearson")

# Log transformation of the response variable
ln_interest_rate <- log(interest_rate)

# After transformation
plot(leverage, ln_interest_rate, main = "After Log Transformation",
     xlab = "Leverage Ratio", ylab = "ln(Interest Rate)", pch = 19)
abline(lm(ln_interest_rate ~ leverage), col = "blue", lwd = 2)
cor(leverage, ln_interest_rate, method = "pearson")
cor.test(leverage, ln_interest_rate, method = "pearson")
```

SPEARMAN RANK CORRELATION — CLAIM & PAYMENT

Ranking observations and measuring the relationship between Claim and Payment

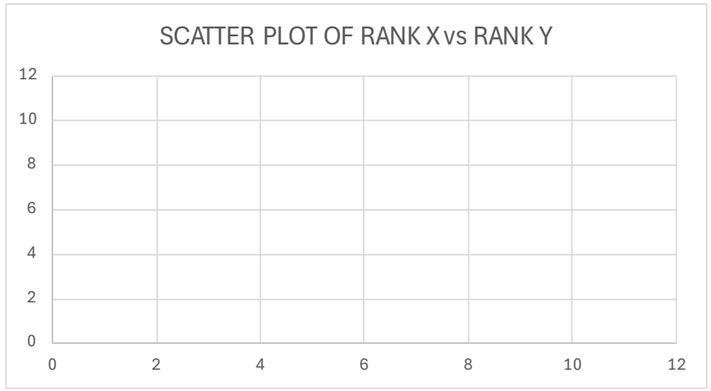
1. DATA AND RANK CALCULATIONS

Claim (X)	Payment (Y)	Rank X	Rank Y	d = Rx – Ry	d^2
2.10	2.18	1	2	-1	1
2.40	2.06	2	1	1	1
2.50	2.54	3	3	0	0
3.20	2.61	4	4	0	0
3.60	3.67	5	6	-1	1
3.80	3.25	6	5	1	1
4.10	4.02	7	8	-1	1
4.20	3.71	8	7	1	1
4.50	4.38	9	9	0	0
5.00	4.45	10	10	0	0
TOTAL				**0**	**6**

2. SPEARMAN CORRELATION FORMULA

$$\rho = 1 - [6\,\Sigma d^2] \,/\, [n(n^2 - 1)]$$

3. SCATTER PLOT



4. SUMMARY STATISTICS

Number of observations (n)	10
Sum of squared rank differences (Σd^2)	6
Spearman ρ — formula	0.9636
Spearman ρ — Excel CORREL	0.9636

5. EXCEL FUNCTIONS

Rank X	RANK.AVG(NUMBER,RANGE,ORDER)
Rank Y	RANK.AVG(NUMBER,RANGE,ORDER)
Spearman ρ	CORREL(RANK X, RANK Y)

5. INTERPRETATION

Spearman's rank correlation is ρ = 0.9636, indicating a very strong positive monotonic relationship between CLAIM and PAYMENT. Higher claim values tend to be associated with higher payment values. There are no tied values in either variable, so the shortcut formula and Excel's CORREL function applied to the ranks give the same result. Correlation does not establish causation.

6. R CODE

```
claim <- c(2.1, 2.4, 2.5, 3.2, 3.6, 3.8, 4.1, 4.2, 4.5, 5.0)
payment <- c(2.18, 2.06, 2.54, 2.61, 3.67, 3.25, 4.02, 3.71, 4.38, 4.45)

# Spearman rank correlation
cor(claim, payment, method = "spearman")
cor.test(claim, payment, method = "spearman", exact = TRUE)

# Calculate ranks and rank differences
rank_x <- rank(claim)
rank_y <- rank(payment)
d <- rank_x - rank_y
d_squared <- d^2
data.frame(claim, payment, rank_x, rank_y, d, d_squared)
sum(d_squared)

# Scatter plot
plot(claim, payment, main = "Claim vs Payment",
     xlab = "Claim (X)", ylab = "Payment (Y)", pch = 19)
```

SPEARMAN RANK CORRELATION — CLAIM & PAYMENT

Ranking observations and measuring the relationship between Claim and Payment

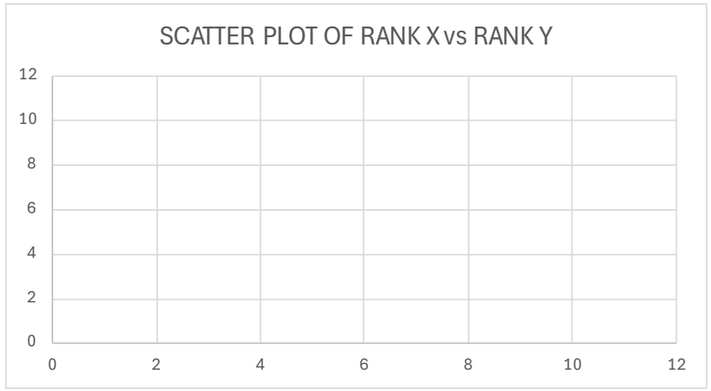
1. DATA AND RANK CALCULATIONS

Leverage Ratio(X)	Interest Rate (Y)	Rank X	Rank Y	d = Rx – Ry	d^2
0.10	2.80	1	1	0	0
0.40	3.40	2	2	0	0
0.50	3.50	3	3	0	0
0.80	3.60	4	4	0	0
1.00	4.60	5	5	0	0
1.80	6.30	6	6	0	0
2.00	10.20	7	7	0	0
2.50	19.70	8	8	0	0
2.80	31.30	9	9	0	0
3.00	42.90	10	10	0	0
TOTAL				**0**	**0**

2. SPEARMAN CORRELATION FORMULA

$$\rho = 1 - [6\,\Sigma d^2] \,/\, [n(n^2 - 1)]$$

3. SCATTER PLOT



4. SUMMARY STATISTICS

Number of observations (n)	10
Sum of squared rank differences (Σd^2)	0
Spearman ρ — formula	1.0000
Spearman ρ — Excel CORREL	1.0000

5. EXCEL FUNCTIONS

Rank X	RANK.AVG(NUMBER,RANGE,ORDER)
Rank Y	RANK.AVG(NUMBER,RANGE,ORDER)
Spearman ρ	CORREL(RANK X, RANK Y)

5. INTERPRETATION

Spearman's rank correlation is ρ = 0.9636, indicating a very strong positive monotonic relationship between CLAIM and PAYMENT. Higher claim values tend to be associated with higher payment values. There are no tied values in either variable, so the shortcut formula and Excel's CORREL function applied to the ranks give the same result. Correlation does not establish causation.

6. R CODE

```
claim <- c(2.1, 2.4, 2.5, 3.2, 3.6, 3.8, 4.1, 4.2, 4.5, 5.0)
payment <- c(2.18, 2.06, 2.54, 2.61, 3.67, 3.25, 4.02, 3.71, 4.38, 4.45)

# Spearman rank correlation
cor(claim, payment, method = "spearman")
cor.test(claim, payment, method = "spearman", exact = TRUE)

# Calculate ranks and rank differences
rank_x <- rank(claim)
rank_y <- rank(payment)
d <- rank_x - rank_y
d_squared <- d^2
data.frame(claim, payment, rank_x, rank_y, d, d_squared)
sum(d_squared)

# Scatter plot
plot(claim, payment, main = "Claim vs Payment",
     xlab = "Claim (X)", ylab = "Payment (Y)", pch = 19)
```

KENDALL'S TAU WITHOUT TIES — CLAIM & PAYMENT

Counting concordant and discordant pairs using pairwise difference products

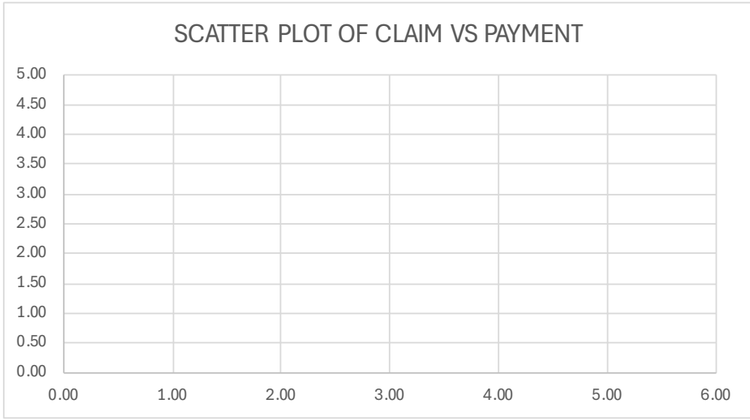
WHAT IS CONCORDANT AND DISCORDANT?

Concordant pair (C)	X and Y move in the same direction across a pair: both increase or both decrease.
Discordant pair (D)	X and Y move in opposite directions across a pair: one increases while the other decreases.
Pairwise product	(Xi − Xj)(Yi − Yj): positive = concordant; negative = discordant; zero = tied pair.

1. DATA AND PAIRWISE COMPARISON PRODUCTS

Obs.	CLAIM (X)	PAYMENT (Y)	VS1	VS2	VS3	VS4	VS5	VS6	VS7	VS8	VS9
1	2.10	2.18									
2	2.40	2.06	-0.0360								
3	2.50	2.54	0.1440	0.0480							
4	3.20	2.61	0.4730	0.4400	0.0490						
5	3.60	3.67	2.2350	1.9320	1.2430	0.4240					
6	3.80	3.25	1.8190	1.6660	0.9230	0.3840	-0.0840				
7	4.10	4.02	3.6800	3.3320	2.3680	1.2690	0.1750	0.2310			
8	4.20	3.71	3.2130	2.9700	1.9890	1.1000	0.0240	0.1840	-0.0310		
9	4.50	4.38	5.2800	4.8720	3.6800	2.3010	0.6390	0.7910	0.1440	0.2010	
10	5.00	4.45	6.5830	6.2140	4.7750	3.1120	1.0920	1.4400	0.3870	0.5920	0.0350

2. SCATTER PLOT



3. HOW TO DETERMINE CONCORDANT AND DISCORDANT

If product > 0	The pair is concordant (C).
If product < 0	The pair is discordant (D).
If product = 0	The pair is tied and is not counted as C or D.

4. Formula

$$\tau_B = \frac{(n_c - n_d)}{n(n-1)/2}$$

5. KENDALL'S TAU-a (NO TIES)

n (number of observations)	10	
Number of Unique Pairs to be Compared	45	*n(n − 1) / 2*
C (concordant pairs)	37	*Count of products greater than 0: COUNTIF($D$15:$L$20,">0")*
D (discordant pairs)	2	*Count of products less than 0: COUNTIF($D$15:$L$20,"<0")*
Kendall tau-a	0.866667	

6. INTERPRETATION

Kendall's tau-a = 0.8667. This indicates a strong positive association between CLAIM and PAYMENT. There are 42 concordant pairs and 3 discordant pairs out of 45 total pairs. Higher claim values generally tend to be associated with higher payment values. Correlation describes association and does not establish causation.

7. R CODE TO VERIFY

```
claim <- c(2.1, 2.4, 2.5, 3.2, 3.6, 3.8, 4.1, 4.2, 4.5, 5.0)
payment <- c(2.18, 2.06, 2.54, 2.61, 3.67, 3.25, 4.02, 3.71, 4.38, 4.45)

# Kendall correlation coefficient
cor(claim, payment, method = "kendall")
cor.test(claim, payment, method = "kendall", exact = TRUE)

# Count concordant and discordant pairs manually
n <- length(claim); C <- 0; D <- 0
for (i in 1:(n - 1)) {
  for (j in (i + 1):n) {
    product <- (claim[i] - claim[j]) * (payment[i] - payment[j])
    if (product > 0) C <- C + 1
    if (product < 0) D <- D + 1
  }
```

```
}
tau_a <- (C - D) / (n * (n - 1) / 2)
C; D; tau_a

# Scatter plot
plot(claim, payment, main = "Claim vs Payment", xlab = "Claim (X)", ylab = "Payment (Y)", pch = 19)
```

SPEARMAN RANK CORRELATION WITH TIES

How to rank repeated values and interpret the relationship

WHAT DOES 'TIES' MEAN?

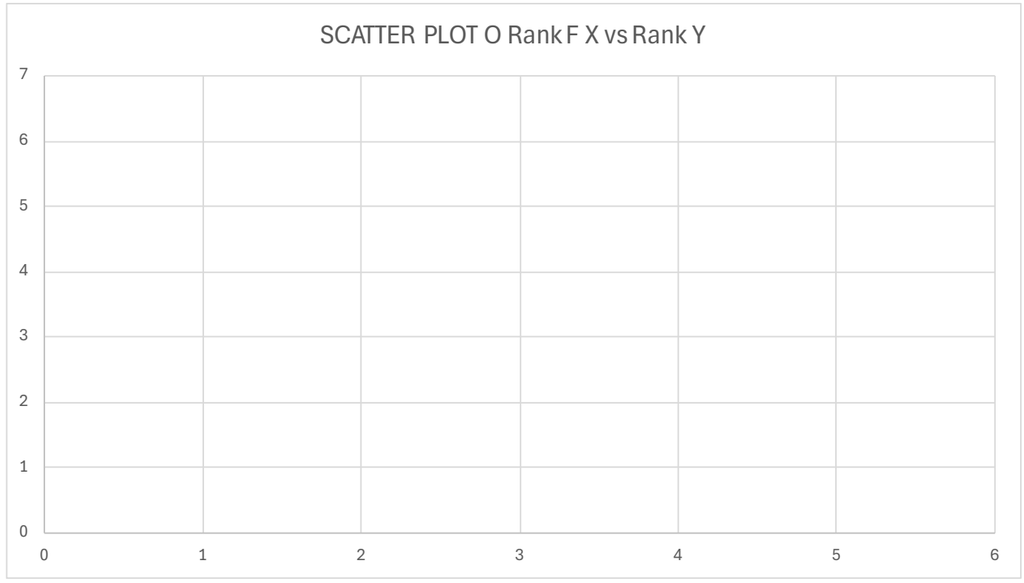
Ties happen when two or more observations have the same value. Example: X = 50 appears twice. Both observations receive the average of the ranks they would occupy.

1. DATA AND RANKS

Obs.	X	Y	Rank X	Rank Y	d = Rx – Ry	d^2
1	30	200	3	2.5	0.5	0.25
2	50	400	5.5	6	-0.5	0.25
3	50	300	5.5	4.5	1	1
4	20	200	2	2.5	-0.5	0.25
5	10	100	1	1	0	0
6	40	300	4	4.5	-0.5	0.25
TOTAL					**0**	**2**

Meaning **d** is the difference between the two ranks; **d^2** is that difference squared.

2. SCATTER PLOT



3. HOW TO ASSIGN RANKS WHEN VALUES REPEAT

Rank X: value 50 appears twice	It would occupy ranks 5 and 6. Average rank = (5 + 6) / 2 = 5.5, so both get rank 5.5.
Rank Y: value 200 appears twice	It would occupy ranks 2 and 3. Average rank = (2 + 3) / 2 = 2.5, so both get rank 2.5.
Rank Y: value 300 appears twice	It would occupy ranks 4 and 5. Average rank = (4 + 5) / 2 = 4.5, so both get rank 4.5.
Excel function	RANK.AVG(NUMBER,RANGE,ORDER)

4. TIE CORRECTION TERMS

Variable	Repeated value	Frequency t	$CF = (t^3 - t)/12$	CF	
X	50	2	0.5	CF_X	0.5
Y	200	2	0.5	CF_Y	1
Y	300	2	0.5		

5. CALCULATING SPEARMAN'S RANK CORRELATION WITH TIES

$$\rho = 1 - \frac{6\left(\sum d_i^2 + CF_x + CF_y\right)}{n(n^2 - 1)}$$

n (number of observations)	6	Excel function: CORREL(RANK X, RANK Y)
Spearman rho (correlation of ranks)-Formula	0.900	Excel's built-in function use average ranks for tied observations and calculate the Pearson correlation between the ranks.
Spearman rho (correlation of ranks)-Excel	0.9404	

6. INTERPRETATION

There is a very strong positive monotonic relationship between X and Y

7. R CODE

```
x <- c(30, 50, 50, 20, 10, 40)
y <- c(200, 400, 300, 200, 100, 300)
cor(x, y, method = "spearman")
# Expected: approximately 0.769 (correlation of average ranks)
cor.test(x, y, method = "spearman", exact = FALSE)
```

KENDALL'S TAU-b WITH TIES

Counting concordant, discordant and tied pairs

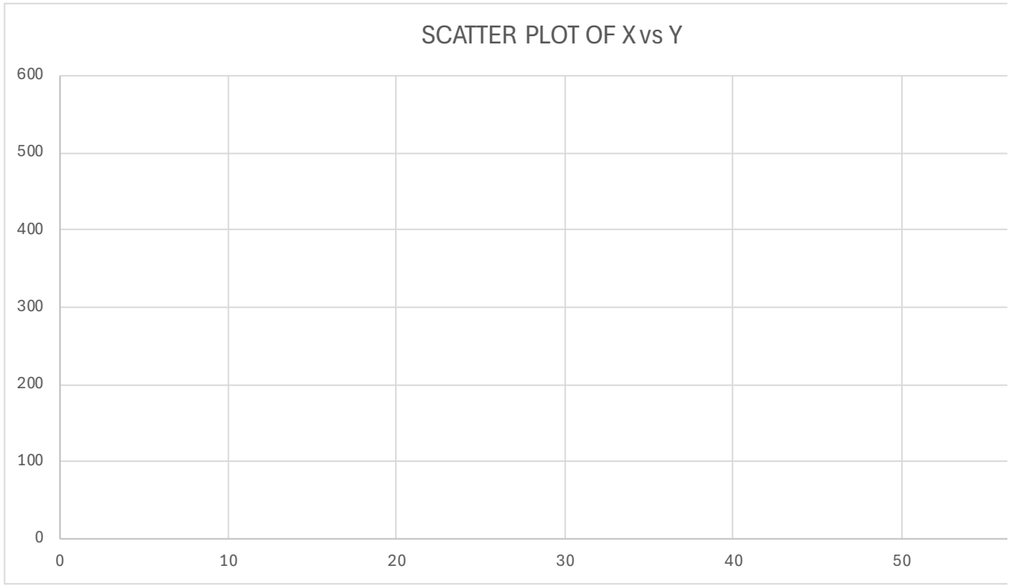
WHAT IS CONCORDANT AND DISCORDANT?

Concordant pair (C)	X and Y move in the same direction across a pair: both increase or both decrease.
Discordant pair (D)	X and Y move in opposite directions across a pair: one increases while the other decreases.
Pairwise product	(Xi − Xj)(Yi − Yj): positive = concordant; negative = discordant; zero = tied pair.

1. DATA

Obs.	X	Y	VS1	VS2	VS3	VS4	VS5
1	10	100					
2	20	200	1000				
3	20	300	2000	0			
4	30	300	4000	1000	0		
5	40	400	9000	4000	2000	1000	
6	50	500	16000	9000	6000	4000	1000

2. SCATTER PLOT



3. HOW TO DETERMINE CONCORDANT AND DISCORDANT

If product > 0	The pair is concordant (C)
If product < 0	The pair is discordant (D)
If product = 0	The pair is tied, and is not counted as C or D.

4. TIE COUNTS

Variable	Repeated value	Frequency, t	T = t(t−1)/2	T
X	20	2	1	T_X
Y	300	3	3	T_Y

5. KENDALL'S TAU-b WITH TIES

$$\tau_B = \frac{(C-D)}{\sqrt{(C+D+T_X)(C+D+T_Y)}} \qquad nCr = \frac{n(n-1)}{2}$$

n (number of observations)	6	
Number of combinations	15	The number of unique pairs to be compared
C (concordant pairs)	13	Excel function: COUNTIF(RANGE,">0")
D (discordant pairs)	0	Excel function: COUNTIF(RANGE,"<0")
Tx (ties in X)	1	
Ty (ties in Y)	3	
Kendall tau-b	0.8686	

6. INTERPRETATION

The coefficient is close to +1, indicating a very strong positive association in this dataset: larger X values tend to occur with larger Y values. This describes association, not causation.

7. R CODE TO VERIFY

```
x <- c(10, 20, 20, 30, 40, 50)
y <- c(100, 200, 300, 300, 400, 500)
cor(x, y, method = "kendall")
cor.test(x, y, method = "kendall", exact = FALSE)
```